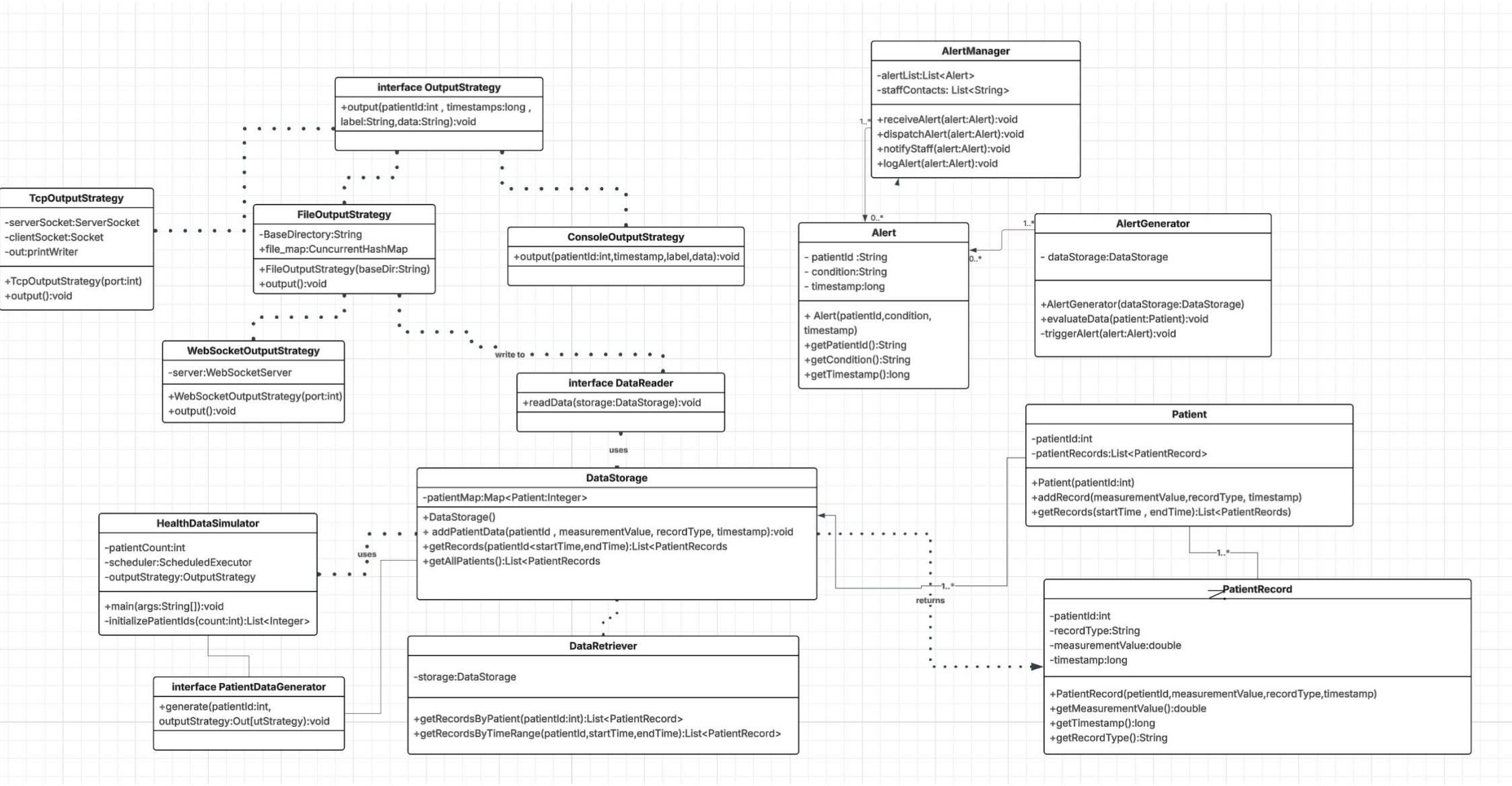
interface OutputStrategy
+output(patientId:int , timestamps:long , label:String,data:String):void
AlertManager
-alertList:List<Alert>
-staffContacts: List<String>
+receiveAlert(alert:Alert):void
+dispatchAlert(alert:Alert):void
+notifyStaff(alert:Alert):void
+logAlert(alert:Alert):void
1..*
0..*
TcpOutputStrategy
-serverSocket:ServerSocket
-clientSocket:Socket
-out:printWriter
+TcpOutputStrategy(port:int)
+output():void
FileOutputStrategy
-BaseDirectory:String
+file_map:CuncurrentHashMap
+FileOutputStrategy(baseDir:String)
+output():void
ConsoleOutputStrategy
+output(patientId:int,timestamp,label,data):void
Alert
- patientId :String
- condition:String
- timestamp:long
+ Alert(patientId,condition, timestamp)
+getPatientId():String
+getCondition():String
+getTimestamp():long
1..*
0..*
AlertGenerator
- dataStorage:DataStorage
+AlertGenerator(dataStorage:DataStorage)
+evaluateData(patient:Patient):void
-triggerAlert(alert:Alert):void
WebSocketOutputStrategy
-server:WebSocketServer
+WebSocketOutputStrategy(port:int)
+output():void
write to
interface DataReader
+readData(storage:DataStorage):void
uses
Patient
-patientId:int
-patientRecords:List<PatientRecord>
+Patient(patientId:int)
+addRecord(measurementValue,recordType, timestamp)
+getRecords(startTime , endTime):List<PatientReords)
DataStorage
-patientMap:Map<Patient:Integer>
+DataStorage()
+ addPatientData(patientId , measurementValue, recordType, timestamp):void
+getRecords(patientId<startTime,endTime):List<PatientRecords
+getAllPatients():List<PatientRecords
HealthDataSimulator
-patientCount:int
-scheduler:ScheduledExecutor
-outputStrategy:OutputStrategy
+main(args:String[]):void
-initializePatientIds(count:int):List<Integer>
uses
1..*
returns
1..*
PatientRecord
-patientId:int
-recordType:String
-measurementValue:double
-timestamp:long
+PatientRecord(petientId,measurementValue,recordType,timestamp)
+getMeasurementValue():double
+getTimestamp():long
+getRecordType():String
DataRetriever
-storage:DataStorage
+getRecordsByPatient(patientId:int):List<PatientRecord>
+getRecordsByTimeRange(patientId,startTime,endTime):List<PatientRecord>
interface PatientDataGenerator
+generate(patientId:int, outputStrategy:Out[utStrategy):void